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Deteksi Dini Retakan Tanah Longsor
berbasis Crowdsourcing & Edge AI

Jaweed & Tim

IYREF 2026 — Semi-Final

Ancaman longsor laten: 14 dari 24 desa di Jenangan, Ponorogo berada di zona kerentanan tanah bergerak.

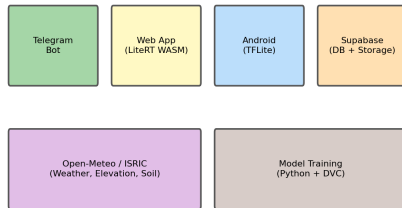
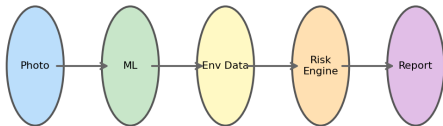
Tantangan:

- Tidak ada sistem peringatan dini berbasis visual
- Warga baru tahu kalo longsor sudah terjadi
- Ahli geologi terbatas, tidak bisa monitor 24/7

Dampak:

- Evakuasi terlambat
- Kerugian harta benda
- Risiko korban jiwa tiap musim hujan

Crowdsourcing + Edge AI: Warga cukup foto retakan tanah dari HP —
ML model menganalisis langsung di perangkat, tanpa perlu cloud.



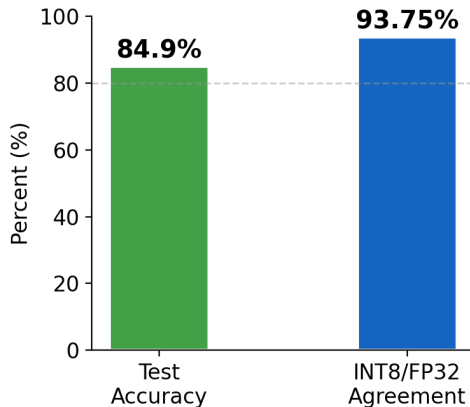
Edge Inference — Satu Model, Dua Platform

Android (Kotlin)

- TFLite Interpreter 2.16.1 (4 thread)
- Resize bilinear 224×224
- Threshold: 40% → AMAN
- Delta OTA model updates

Web App (TypeScript)

- LiteRT WASM @litertjs/core
- Resize high-quality 224×224
- Threshold: 50% → TIDAK PASTI
- PWA cache offline-ready



Identical pipeline:

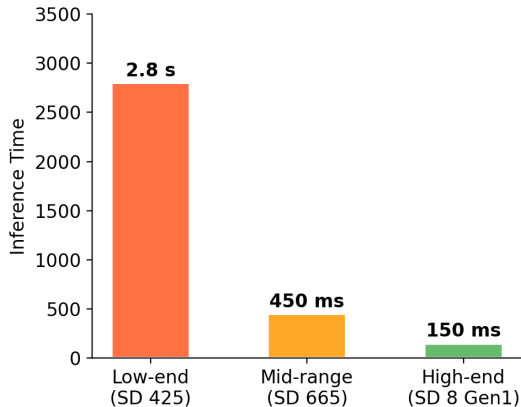
- Model:

Model Machine Learning

Parameter	Value
Architecture	MobileNetV2
Input	224×224×3 uint8
Parameters	~2.26M
Quantization	INT8 PTQ
File size	2.6 MB
Classes	3 (AMAN, WASPADA, BAHAYA)

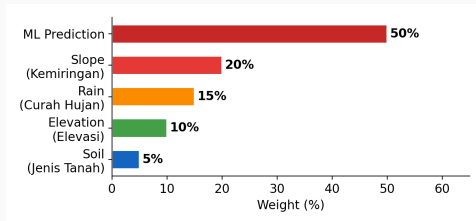
Accuracy: 84.9% test accuracy

FP32 agreement: 93.75%



Multi-Factor Risk Engine

ML saja tidak cukup — risiko longsor dipengaruhi banyak faktor:



Bobot:

- ML Prediction: **50%**
- Slope: **20%** — kemiringan lereng
- Rain: **15%** — curah hujan real-time
- Elevation: **10%** — ketinggian
- Soil: **5%** — jenis tanah

Graceful degradation: Jika API lingkungan gagal, bobot didistribusi ulang ke faktor lain.

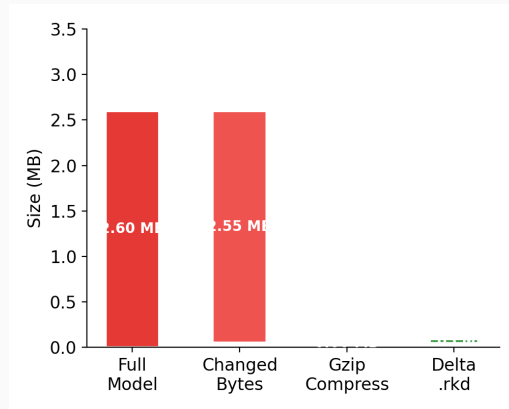
Delta OTA — 98% Bandwidth Savings

Masalah: $10.000 \text{ user} \times 2.6 \text{ MB} = 26 \text{ GB}$ bandwidth per rilis model.

Solusi: Byte-level delta patch

- Hanya kirim byte yang berubah antar versi
- Format .rkd: gzip + region patches
- Gates: savings > 50%, size match, TFLite validation

Real result:	Full model	2.6 MB
	Delta .rkd	47 KB
	Savings	98.2%



Bot self-hosted untuk warga yang tidak menggunakan Android app.

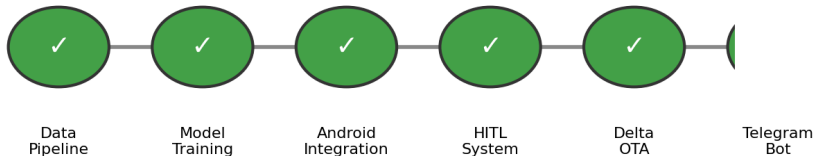
Fitur:

- Kirim foto → ML langsung
- Kirim lokasi → analisis lingkungan
- Sequential flow (foto dulu / lokasi dulu)
- Laporan multifaktor lengkap
- Notifikasi admin kalo BAHAYA

Teknis:

- Python + python-telegram-bot
- Docker + uv build
- Polling mode (no SSL needed)
- Rate limited (10 req/60s)
- Supabase optional

Progress & Roadmap



Done

- Data pipeline
- Model training v3a
- Android app
- Web app
- Multi-factor risk
- Telegram bot
- Delta OTA

In Progress

- Deployment produksi
- Admin dashboard
- HITL loop

Next

- Proximity alert
- Kearifan lokal
- Bimtek warga

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One model, two platforms, zero excuses.

 github.com/jaweed3/retakId